

Attention Mechanisms and Transformer Architectures for Neural Language Understanding

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ABSTRACT

This paper investigates self-attention mechanisms and multi-head attention layers in Transformer neural networks. We demonstrate how feedforward networks and positional encodings enable parallel processing of sequential text data. Our experimental results on large-scale natural language processing benchmarks indicate superior BLEU scores and faster training convergence compared to traditional recurrent neural networks. Furthermore, scaling laws indicate continuous performance improvements with larger parameter sizes.

Keywords: Transformer, Self-Attention, Natural Language Processing, Deep Learning, Neural Networks

1. Introduction

Sequence transduction models have traditionally relied on complex recurrent or convolutional neural networks. Recurrent architectures inherently preclude parallelization within training examples. The Transformer model bypasses recurrence entirely, relying exclusively on an attention mechanism to draw global dependencies between input and output tokens.

2. Architecture & Methods

The architecture follows an encoder-decoder structure. The encoder is composed of a stack of identical layers, each containing multi-head self-attention and position-wise feed-forward networks. Layer normalization and residual connections are applied after each sub-layer to stabilize gradient flow.

3. Results & Evaluation

Experiments on machine translation and language understanding benchmarks show that the Transformer achieves state-of-the-art results while requiring significantly less computation time to train.

4. Conclusion

Self-attention provides a scalable, computationally efficient paradigm for representation learning across natural language processing and multimodal foundation models.